



How emissions trading system inhibits carbon emission intensity: evidence from global carbon markets

Wenzhi Wang^a, Sijia Fan^a, Yong Hu^{b,*} 

^a School of Economics, Tianjin Normal University, Tianjin, China

^b School of Economics, Zhejiang Gongshang University, Zhejiang, China

ARTICLE INFO

Keywords:

Emission trading system
Carbon emission intensity
Multi-period difference-in-difference
Carbon Leakage

ABSTRACT

As a market-based climate governance instrument integrating both incentives and constraints, the Emissions Trading System (ETS) has attracted considerable attention for its emission reduction effectiveness. Exploiting the global variation in ETS adoption across countries, this study employs a multi-period difference-in-difference model to examine the mitigation effects of ETS implementation on carbon emission intensity across diverse countries and to identify the underlying mechanisms. The empirical results demonstrate that ETS implementation significantly reduces carbon emission intensity in participating countries, and these findings remain robust after a series of robustness checks. The emission reduction effect of ETS operates primarily through optimizing energy consumption structure and stimulating green technology innovation. Financial development and the maturity of the carbon market can strengthen the mitigating effect of the ETS on carbon emission intensity. However, we also find that ETS implementation may induce global carbon leakage, and this trade-off effect exerts adverse impacts on global carbon abatement. These findings provide important implications for strengthening international carbon market cooperation, improving carbon market linkages, and advancing the development of a globally integrated carbon market.

1. Introduction

The 2024 Emissions Gap Report by the United Nations Environment Program pointed out that, without raising and immediately implementing the next round of Nationally Determined Contributions (NDCs), global temperatures will rise by 2.6–3.1 °C this century and the climate objectives of the Paris Agreement will become unattainable within a few years, with devastating consequences for humanity and the economy. Achieving global emission reduction targets requires not only updating national NDCs but also accurately assessing the emission reduction effectiveness of existing mainstream climate policy instruments [1]. In response to unprecedented levels of global greenhouse gas emissions, countries have introduced a variety of climate policies. The OECD's Climate Actions and Policies Measurement Framework (CAPMF) has tracked 130 climate policies in total. Data shows that the number of policies implemented across the four key sectors (electricity, industry, buildings and transport) has increased continuously from 1998 to 2022. The average number of climate policies adopted by countries was <1 in 1998, rising to between 4 and 8 by 2022. The Emissions Trading System (ETS) and the Carbon Tax are key climate policy instruments that have

been adopted by 73 countries worldwide. These instruments cover approximately 23% of global greenhouse gas (GHG) emissions, with the ETS accounting for 18%. Most policymakers and scholars consider market-based ETS to be the most cost-effective way of reducing emissions [2]. However, according to the World Bank's 2023 Carbon Pricing Report, global carbon prices vary widely—from \$1 to \$139 per tonne—with 46% of countries pricing carbon below \$10 per tonne, far short of the \$40–80 range recommended for meeting the Paris Agreement's temperature targets. Such low prices may undermine the incentive for emission reductions. Thus, empirically assessing the mitigation effectiveness of ETS from a global perspective and elucidating their underlying mechanisms is essential for improving carbon market design and advancing the goals of the Paris Agreement.

Whether ETS can effectively curb the increase in greenhouse gas emissions is still highly controversial. Some scholars argue that ETS fail to achieve substantive emission reductions. For instance, Chen et al. [3] find that ETS has restrained the green innovation of enterprises in China. The trading of surplus permits by low-emission firms to high bidders merely redistributes emission allowances without lowering total global emissions [4]. Moreover, free allocation of quotas can result in

* Corresponding author.

E-mail address: huyong@zjgsu.edu.cn (Y. Hu).

<https://doi.org/10.1016/j.sfr.2026.102156>

Received 18 May 2026; Received in revised form 5 August 2026; Accepted 14 September 2026

Available online 17 September 2026

2666-1888/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

inefficient incentives across industries [5], while corporate arbitrage may cause carbon leakage to regions with lax environmental regulations [6]. In contrast, other studies argue that ETS can effectively mitigate carbon emissions [7–9]. On the one hand, ETS impose both explicit and implicit costs on firms—such as those related to carbon accounting, reporting, and environmental governance—thereby restraining the output of emission-intensive enterprises [10]. The magnitude of this cost-induced suppression effect depends on factors including carbon market liquidity [11], the mandatory nature of emission reductions [12], and the allocation method of emission allowances [13–15]). This cost effect tends to be more pronounced in mature ETS markets located in economically advanced regions [16]. On the other hand, the innovation incentive effect of ETS (Porter Hypothesis) strengthens the benefit expectations of technological innovation through carbon price signals and market trading mechanisms. The benefit expectations will drive companies to increase R&D investment in low-carbon technologies [17], thereby effectively promote carbon emission reduction of enterprises [18–21], and reduce their carbon emission intensity [22–24]. However, the incentive efficacy of ETS may be undermined by inadequate market liquidity or suboptimal policy design [25,26]. In regions with underdeveloped market mechanisms, emission reductions often depend more on administrative interventions than on market-driven incentives [21,27].

Overall, the divergence in these opinions can be attributed to the following reasons. First, most existing studies on the carbon mitigation effects of ETS focus on developed countries such as those in the EU and the United States, where the implementation of ETS has generally led to significant emission reductions. In contrast, developing countries often exhibit underdeveloped carbon market infrastructures and relatively low carbon prices, casting doubt on the effectiveness of ETS in such contexts [1]. Second, variations in carbon pricing levels influence the strength and transmission of price signals, which in turn moderate the emission reduction impact of ETS. Third, ETS implementation may induce “carbon leakage,” whereby emissions are shifted to regions with laxer regulations, resulting in a trade-off pattern of global carbon emissions and thereby undermining the effectiveness of carbon reduction efforts. For instance, the European Union introduced the Carbon Border Adjustment Mechanism (CBAM) in 2023 to address carbon leakage associated with the EU ETS. Consequently, it is essential to consider the trade-off effects of ETS when assessing its overall effectiveness.

By addressing the above issues, this paper contributes to the existing literature as follows. First, this study selects the top 120 economies ranked by global economic development level in 2022. After excluding economies with missing values for control variables, a valid sample of 102 economies is retained, including 42 developed economies and 60 developing economies. The sample covers approximately 87.31% of global carbon emissions in 2022. Among them, 29 developed economies and 9 developing economies have implemented Emissions Trading Systems (ETS). A multi-period Difference-in-Differences (DID) model is also adopted to verify the effect of ETS on local carbon emission intensity and examine the underlying mechanisms.

Second, we investigate the mediating mechanisms through which ETS reduces carbon emission intensity, with a particular focus on energy consumption structure and green technological innovation. Furthermore, we identify and examine the moderating factors that shape the emission reduction effects of ETS. The results indicate that both financial development and carbon market maturity significantly strengthen the emission reduction effects of ETS. In addition, regional natural geographical factors, including the cumulative factor, deposition factor, and carbon sequestration factor, also influence the emission reduction effects of ETS. The heterogeneity analysis provides an explanation for why some countries with relatively low carbon emissions and carbon emission intensity have not implemented ETS. Moreover, the subgroup regression results demonstrate that these geographical factors significantly affect the emission reduction performance of ETS.

Third, by employing a Multi-Regional Input–Output (MRIO) model to measure carbon transfers among 68 economies, we find that ETS significantly contributes to global carbon leakage and generates a notable trade-off between emission intensity reduction and carbon leakage. The disparities in carbon market maturity across countries incentivize the relocation of pollution-intensive production activities toward jurisdictions with relatively weaker environmental regulations, thereby undermining the effectiveness of global emission reduction efforts. These findings highlight the importance of strengthening international carbon market linkages [28] and establishing a more integrated global carbon market framework to achieve reductions in carbon emission intensity while simultaneously curbing global carbon leakage. In response to this challenge, international coordination on carbon markets has recently gained increasing attention. On November 8, 2025, the European Union, together with China, the United Kingdom, Canada, Chile, France, Germany, Mexico, Zambia, Oman, and other countries, officially joined the Open Coalition on Compliance Carbon Markets initiated by Brazil. This initiative aims to establish a transnational cooperation framework that coordinates carbon pricing mechanisms, emissions trading systems, and related climate policies among participating countries, thereby promoting an interconnected, transparent, and credible global compliance carbon market network. The findings of this study provide theoretical support for such international coordination efforts and offer empirical insights into the design of global carbon market governance mechanisms.

Section 2 develops the theoretical framework and proposes hypotheses regarding the effects of ETS on carbon emission intensity. Section 3 describes the empirical strategy, including the identification framework employed to estimate the causal effects of ETS on carbon emission intensity and the corresponding statistical methods. Section 4 presents the main empirical results, followed by a series of robustness checks and further analyses of the underlying mediating and moderating mechanisms. Section 5 examines embodied carbon transfers and investigates the potential trade-off between the emission reduction effects of ETS and carbon leakage. Finally, Section 6 concludes by summarizing the main findings, discussing policy implications, and highlighting research limitations.

2. Research hypotheses

2.1. Impacts of ETS on carbon emission intensity

According to property rights theory, clearly defined property rights enable optimal resource allocation through market mechanisms. The carbon market embodies this principle by transforming carbon emission rights, which were previously regarded as public environmental goods, into tradable scarce resources, thereby internalizing the negative externalities of corporate pollution as part of firms’ operational costs [29]. Such a cost-following effect increases the compliance costs of firms participating in ETS, thereby encouraging them to reduce the production of carbon-intensive goods, increase the use of renewable energy, and promote the production of low-carbon products. These adjustments ultimately contribute to a decline in national carbon emission intensity. Specifically, ETS implementation will promote the transition toward renewable energy. By setting carbon emission caps, ETS increases the cost of fossil fuel consumption [30], thereby incentivizing firms to invest in or adopt renewable energy sources [31,32]. For example, trading mechanisms such as Certified Emission Reductions (CERs) provide additional revenue streams for renewable energy projects. As the share of renewable electricity generation increases, regional power systems become less dependent on fossil fuels, leading to a decline in emissions from electricity consumption [33]. Given that the power sector represents a major source of carbon emissions in many economies, the expansion of renewable energy plays a critical role in lowering regional carbon emission intensity. Consequently, stringent ETS regulations can facilitate a shift from fossil fuels to renewable energy [34], thereby

reducing overall carbon emission intensity.

Meanwhile, as a market-based environmental policy instrument with both regulatory and incentive functions, ETS raises compliance costs, compelling firms to reduce emissions through renewable energy inputs and lower-intensity production methods [35]. Simultaneously, green technological progress is the main channel for reducing carbon emission intensity [36–38], ETS motivates firms to invest in green technological innovation to gain surplus returns in the carbon market [39–43], thus generating an “innovation incentive effect” [44]. The higher the carbon price, the stronger the incentive for firms to pursue technological advancement [14]. Specifically, when the emission reduction benefits derived from such innovation are monetized through carbon trading, firms’ motivation for further innovation is enhanced, creating a positive feedback loop that accelerates their low-carbon transition [45]. Furthermore, advances in green technologies can diffuse across firms and industries through knowledge spillovers, thereby reducing carbon intensity throughout the broader industrial chain. At the regional level, both the cost-following and innovation incentive effects contribute to a reduction in aggregate carbon emission intensity. Accordingly, we propose the following hypothesis:

H1: *ETS significantly reduces carbon emission intensity in implementing regions by optimizing energy consumption structures and promoting green technological innovation.*

2.2. Moderating factors

Financial development of countries will influence the effectiveness of ETS in reducing carbon emissions. On the one hand, financial development enhances ETS performance through improved market liquidity, optimized resource allocation, and innovation in financial instruments. A more developed financial system expands the market scale of carbon financial products and increases transaction activity, thereby enabling carbon prices to better reflect environmental costs [46]. This, in turn, incentivizes firms to lower carbon costs through technological innovation, enhancing ETS’s emission reduction impact [47]. As financial support systems improve, capital increasingly flows toward green and high value-added industries. This reallocation promotes resource efficiency and drives low-carbon economic transformation. Furthermore, green financial innovation—such as carbon asset pledge financing (CAPF), carbon delivery guarantee insurance (CDGI), carbon futures and options—diversifies trading mechanisms, reduces emission reduction costs, and strengthens the role of ETS [48]. However, financial development plays a dual role in the emission-reduction mechanism of carbon emission trading systems. In the absence of effective financial regulation, financial development can weaken carbon abatement performance through capital misallocation, distorted carbon pricing [49], speculative disturbances, and regulatory arbitrage [50].

As a market-oriented emission reduction instrument, the core mechanism of the ETS lies in guiding participating enterprises to adjust their production behaviors through price signals, thereby reducing their carbon emissions. Fluctuations in carbon trading prices facilitate the market-based valuation of emission allowances and represent the most direct manifestation of market mechanisms [51]. The carbon price serves as an indicator of the degree of marketization. Higher prices reflect a more mature carbon market with stronger signaling functions [52]. Under higher carbon prices, the economic penalties for excessive emissions become more severe, incentivizing firms to actively pursue emission reduction strategies such as restructuring their energy mix, investing in cleaner production technologies, and adopting more efficient and low-carbon production processes and equipment, thereby curbing emissions at the source. Hence, a mature carbon market can stimulate technological innovation and low-carbon investment by providing appropriate price incentives, channeling more market capital toward emission-reducing firms, and ultimately enhancing the effectiveness of the ETS in reducing carbon emission intensity. Accordingly, the following hypothesis is proposed:

H2: *Both the financial development and the maturity of the carbon market may strengthen the mitigating effect of the ETS on carbon emission intensity.*

2.3. Trade-off effect

Differences in the development of national carbon markets may lead to heterogeneous effects of ETS on carbon emission intensity reduction. In general, well-established and mature carbon markets, such as the EU ETS, tend to generate more significant emission reduction effects [53]. Meng et al. [54] demonstrate that narrowing the gap in emission intensities between developed and developing countries can mitigate global carbon leakage associated with international trade. In contrast, widening disparities in sectoral carbon emission intensities across countries may further promote global carbon leakage, resulting in a “carbon displacement” phenomenon in which emission reductions in one region are offset by increases elsewhere. Specifically, in economies with relatively high carbon market prices, firms face greater environmental compliance costs and may relocate carbon-intensive products and production processes to economies with lower environmental costs through international trade, thereby achieving cost-efficient emission reductions. However, economies with lower carbon prices may experience weakened mitigation effectiveness due to the inflow of carbon-intensive production and net embodied carbon emissions. This trade-off effect induced by the establishment of ETS may undermine the overall effectiveness of global carbon mitigation efforts. Therefore, strengthening international linkages among carbon markets and reducing disparities in carbon prices across countries are essential for mitigating global carbon leakage. Based on this reasoning, we propose the following hypothesis:

H3: *While carbon markets can reduce carbon emission intensity, they may also induce cross-border carbon transfers, creating a trade-off between emission intensity reduction and carbon leakage that undermines global carbon emission reduction efforts.*

In summary, the impact mechanisms of ETS on carbon emission intensity are illustrated in Fig. 1.

3. Methodology and data

3.1. Benchmark regression model

Considering the variation in ETS implementation timing across countries and regions, we select the top 102 economies by economic development level in 2022—including 42 developed and 60 developing countries. Among them, 29 developed and 9 developing countries have adopted ETS (see Appendix 1). Based on this, we construct a multi-period difference-in-differences (DID) model.

$$CEI_{it} = \alpha + \beta \times ETS_{it} + \delta \times X_{it} + \phi_i + \phi_t + \varepsilon_{it} \quad (1)$$

In Eq. (1), i denotes the country, and t denotes the year; CEI_{it} represents the carbon emission intensity of country i in year t ; ETS_{it} is the interaction term of $Treat_i$ and $Post_{it}$, indicating whether country i has implemented ETS in year t and capturing the treatment effect of ETS implementation. $Treat_i$ is a binary variable indicating whether country i is in the treatment group (assigned 1 if ETS was implemented during the sample period; 0 otherwise). while $Post_{it}$ is a time dummy equal to 1 in and after the year of ETS implementation, and 0 otherwise. X_{it} is a series of control variables, ϕ_i and ϕ_t represent country fixed effects and year fixed effects, respectively, and ε_{it} represents a random disturbance term. Robust standard errors are clustered at the country level.

3.2. Mediation effect model

To examine the indirect effects of ETS implementation on carbon emission intensity, A mediation effect model is constructed to explore two potential transmission pathways, which are optimization of the

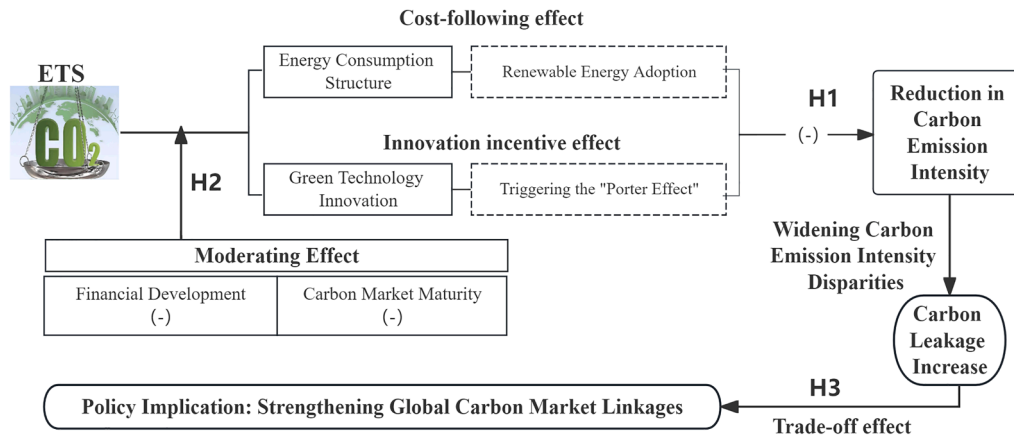


Fig. 1. Impact mechanisms of ETS on carbon emission intensity.

energy consumption structure and promotion of green technological innovation. Since the reduction in carbon emission intensity driven by energy structure optimization and green technological innovation has been confirmed in the majority of relevant studies [31,32,34,36–38], this study adopts a two-stage mediation analysis framework. The model specification is presented in Eq. (2), where $Mediator_{it}$ denotes the mediating variables for country i in year t . All other variables are defined as in Eq. (2).

$$Mediator_{it} = \alpha_0 + \alpha_1 ETS_{it} + \alpha_2 X_{it} + \phi_i + \phi_t + \varepsilon_{it} \quad (2)$$

3.3. Variable explanation

3.3.1. Dependent variable (CEI)

Carbon emission intensity (CEI) is measured as carbon dioxide emissions per unit of gross domestic product (GDP), capturing the degree of decoupling between economic growth and environmental pressure. A decline in CEI indicates improved carbon efficiency, reflecting a reduction in emissions per unit of economic output. In this study, CEI is represented by CO₂ emissions per unit of GDP (in PPP dollars), sourced from the World Bank's World Development Indicators (WDI) database.

3.3.2. Independent variable (ETS)

The implementation of ETS policies is identified based on data from the World Bank's World Development Indicators (WDI) on carbon market adoption across countries. Regions that have established operational carbon markets in a given year are assigned a value of 1 and classified as the treatment group, while those that have not implemented or are merely considering ETS are assigned a value of 0 and serve as the control group. As of March 2025, 38 countries or regions have adopted ETS, including 29 developed and 9 developing economies; the remaining 64 constitute the control group.

3.3.3. Control variables

Drawing on relevant studies [34,55], this study incorporates the following control variables: population size (lnPEO), development level of the tertiary industry (lnSER), development level of the secondary industry (lnIND), development level of the primary industry (lnAGR), electricity coverage rate (lnELE), manufacturing development level (lnMAN), unemployment level (lnUNE), government expenditure level (lnGOV), and trade competitiveness (RCA). Detailed variable definitions are provided in Table 1.

3.3.4. Mediating variables

Energy Consumption Structure (ECS): We use the proportion of renewable energy generation in total electricity generation—based on internationally comparable data from the World Bank's WDI

Table 1
Descriptive statistics.

Variable	Description	Obs.	Mean	Std. Dev.
CEI	Carbon Emission Intensity (kg/USD PPP)	2328	0.279	0.179
ETS	Whether a Carbon Emission Trading Scheme Pilot is Conducted in the Current Year (Yes=1, No=0)	2328	0.232	0.422
lnPEO	Total National Population	2328	15.853	1.803
lnSER	Service Industry Value Added as % of GDP	2328	4.013	0.226
lnIND	Industrial Value Added as % of GDP	2328	3.166	0.095
lnAGR	Agricultural Value Added as % of GDP	2328	1.631	0.741
lnELE	Electrification Rate (Percentage of Population)	2328	4.553	0.155
lnMAN	Manufacturing Share (Percentage of GDP)	2328	2.502	0.550
lnUNE	Total Number of Unemployed as % of Total Labor Force	2328	1.943	0.737
lnGOV	General Government Final Consumption Expenditure as % of GDP	2328	2.806	0.303
RCA	Revealed Comparative Advantage Index	2328	0.904	5.207
ECS	Renewable Energy Power Generation as % of Total Power Generation	2138	32.752	31.144
GTI	Environmental Patents per Capita	1942	8.659	16.354
RENT	Total Natural Resource Rents as % of GDP	2328	6.61	11.346
HEI	Land Area Below 5 Meters Above Sea Level as % of Total Land Area	1936	4.861	10.009
APD	the long-term average depth (in millimeters per year) of annual precipitation within a country	2259	1083.866	793.653
FIN	the ratio of bank capital to total capital	2035	9.624	3.685
C_PRICE	Carbon Trading Prices of Carbon Markets across Countries	2121	4.093	12.527
C_RATIO	Share of Emissions Covered by Carbon Markets across Countries	1993	0.010	0.017

database—as a mediating variable to quantitatively assess the transmission effect of energy structure optimization in reducing carbon emission intensity. This indicator reflects the proportion of electricity generated from renewable sources (e.g., wind power, photovoltaic power, and hydro-power) relative to total electricity output, including that from fossil fuels and nuclear energy.

Green Technology Innovation (GTI): The level of green innovation technology is proxied by the per capita share of environmental patents, derived from the OECD environmental patent database. This indicator is

calculated as the number of environmental patents—covering areas such as renewable energy, pollution control, energy efficiency, and waste management—divided by the total population. It reflects a country's investment and achievements in clean technology R&D and is an explicit indicator of green technology progress. As a mediating variable, it serves to quantify the transmission effect of green technology innovation in reducing carbon emission intensity.

3.3.5. Moderating variables

Financial development (FIN) is proxied by the ratio of bank capital to total capital obtained from the World Bank Global Financial Development Database. This indicator directly reflects the stability and risk-resilience of the financial system. Its magnitude indicates the sustainability of banks' financing support for the real economy and provides a comprehensive measure of the strength and efficiency of the financial system.

Carbon Market Maturity (CMM) is measured by carbon trading prices (C.PRICE) and emission coverage scope (C.RATIO) of national carbon markets. Following Xu and Zhang [56], the carbon trading price reflects the supply-demand relationship and the effectiveness of price signals within a given carbon market, while the emission coverage scope reflects the scale and coverage scope of the carbon market. Higher values of both indicators indicate a higher level of carbon market maturity. The data are obtained from the World Bank's Carbon Pricing Dashboard, which provides information on major carbon markets, including those in the European Union, China, the United States, and Australia. For countries without ETS implementation, both carbon prices and emission coverage are assigned a value of zero, and these countries serve as the control group. For countries that have established ETSs but lack available statistical information, the corresponding observations are treated as missing values.

3.3.6. Heterogeneous characteristic: natural geographic factors

From the perspective of natural geographical factors, regional carbon sequestration capacity is influenced by the accumulation, deposition, and carbon sequestration factors. Regarding the accumulation factor, implementing pollution control projects in high-altitude regions often involves greater technological challenges and higher labor costs. Moreover, relatively harsh natural conditions may reduce the effectiveness of pollution control technologies, thereby weakening the adaptability and effectiveness of ETS [55,57]. In addition, topographical characteristics influence the dispersion of air pollutants by affecting airflow velocity. For example, plateau, mountainous, and hilly terrains can obstruct air circulation, reduce wind speed, and inhibit pollutant dispersion, thereby weakening the effectiveness of ETS implementation [58]. The cumulative factor is proxied by the proportion of land located at elevations below five meters relative to total national territory [59]. The corresponding data are obtained from the WDI database.

Regarding the carbon sequestration factor, countries with a high dependence on natural resources generally exhibit industrial structures that rely heavily on carbon-intensive sectors, such as fossil fuel extraction and energy-intensive manufacturing. Under the pricing mechanism of carbon emissions trading markets, the marginal abatement costs of these industries tend to increase rapidly. To avoid penalties associated with excessive emissions, firms in these economies are more likely to pursue emission reductions through technological upgrading (e.g., cleaner production technologies) and production adjustments (e.g., reducing carbon-intensive capacity). Meanwhile, a higher dependence on natural resources often implies a larger emissions base, which may amplify the cost constraints and strengthen firms' incentives for emission reduction [55]. The carbon sequestration factor is measured using the natural resource dependence indicator (RENT), defined as the ratio of total natural resource rents to GDP in each country. The World Bank's WDI database reports rents from five categories of resources, oil, natural gas, coal, minerals, and forests. This indicator captures the degree of an economy's dependence on carbon-intensive natural resources (e.g.,

fossil fuels).

Regarding the deposition factor, increased precipitation can reduce the accumulation of particulate air pollutants [60]. Specifically, precipitation facilitates the wet deposition of fine particulate matter suspended in the atmosphere, thereby accelerating the removal of pollutants and potentially enhancing the effectiveness of ETS implementation. The deposition factor is measured by the average annual precipitation depth (in millimeters) at the national level, defined as the long-term average annual precipitation within each country. Adequate precipitation can enhance ecosystem carbon sequestration capacity by strengthening natural carbon capture and storage processes, thereby partially offsetting anthropogenic carbon emissions. The corresponding data are obtained from the WDI database.

3.4. Data sources and processing

This study employs national-level panel data during 2000–2022. Key variables—including carbon emission intensity, emissions trading system, population size, industrial structure, electricity coverage, manufacturing development, unemployment rate, government expenditure, energy consumption structure, natural resource dependence, low-altitude development, and financial development—are primarily sourced from the World Bank. Data on green technological innovation are obtained from the OECD, while export competitiveness is calculated using the UN Comtrade database. Table 1 lists the descriptive statistics of the variables.

4. Empirical results

4.1. Benchmark regression results

Table 2 reports the benchmark regression results based on Eq. (1), examining the impact of ETS implementation on carbon emission intensity. Columns 1–4 sequentially incorporate time fixed effects, country fixed effects, and control variables. The results consistently indicate that ETS implementation significantly reduces carbon emission intensity. Compared to the control group, countries adopting ETS exhibit a marked decline in emission intensity, partially supporting Hypothesis 1.

4.2. Parallel trend test

The multi-period DID regression results indicate that ETS implementation significantly reduces carbon emission intensity in the treatment group. However, this effect relies on the parallel trends assumption—that the treatment and control groups exhibit similar pre-policy trends. Following Beck et al. [61], an event study approach is employed to assess the validity of this assumption. The test model is:

$$CEI_{it} = \alpha + \beta \sum_{k=-7}^{k=4} ETS_{it} + \delta X_{it} + \phi_i + \phi_t + \varepsilon_{it} \quad (6)$$

In Eq. (6), k denotes the relative time period of countries

Table 2
Benchmark model estimation results.

	(1) CEI_{it}	(2) CEI_{it}	(3) CEI_{it}	(4) CEI_{it}
ETS	−0.192*** (−10.87)	−0.068*** (−2.87)	−0.124*** (−7.91)	−0.055** (−2.44)
Control	NO	NO	YES	YES
Country fixed	YES	YES	YES	YES
Time fixed	NO	YES	NO	YES
N	2328	2328	2328	2328
R^2	0.709	0.817	0.816	0.859

Note: *, ** and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. t-statistics are reported in parentheses. Robust standard errors are clustered at the country level. The same applies below.

implementing ETS policy, $k < 0$ refers to the years prior to implementation, while $k \geq 0$ represents the implementation year and subsequent periods. To avoid multicollinearity, the period immediately preceding implementation ($k = -1$) is omitted and serves as the reference period. Fig. 2 presents the parallel trend test results at the 95% confidence level. The findings show no significant pre-treatment differences in carbon emission intensity between the treatment and control groups, supporting the validity of the parallel trends assumption in the DID model. Although a declining trend is observed prior to ETS implementation, the coefficients are statistically insignificant. Starting from $k = 0$ onward, the estimated β coefficients are significantly negative and increase in magnitude over time, indicating that the emission-reducing effect of the ETS gradually strengthens after its implementation.

4.3. Placebo test

To mitigate bias from unobservable confounding factors that may undermine the validity of the Difference-in-Differences model, this paper follows Gong et al. [62] and conducts counterfactual analysis by randomly assigning policy pilot status across sampled economies to construct fictitious treatment groups. To rule out disturbances from low-probability events, we repeat this random assignment procedure 500 times and perform regressions accordingly. As shown in Fig. 3, the estimated coefficients β^{false} cluster around zero and follow a normal distribution, with the vast majority statistically insignificant. The vertical dashed line marks the estimated coefficient of the true core explanatory variable. The results indicate that randomly simulated policy shocks exert no significant impact on carbon emission intensity, which further confirms the reliability of the ETS in curbing carbon emission intensity.

4.4. Heterogeneity treatment effect test

The multi-period DID model employed in the baseline regression incorporates multiple treatment initiation points, which helps to mitigate confounding effects arising from other policy shocks and concurrent time trends, thereby enhancing the robustness of the empirical results [63]. However, when the treatment effects of the multi-period DID are different in the group and time dimensions, the estimated average treatment effect may still be biased—even under the parallel trends assumption—and may potentially yield misleading or even reversed causal inferences [64]. To address this concern, the following section conducts a more detailed diagnostic analysis of treatment effect heterogeneity based on the baseline regression model.

Drawing on the group-time average treatment effect method proposed by Callaway and Sant'Anna [65], the multi-period, multi-unit DID average treatment effect method by De Chaisemartin and d'Haultfoeuille [66], the imputation estimator by Borusyak, Jaravel, and Spiess [67], and the event-study estimator with moving interaction-weighted treatment effects by Sun and Abraham [68], we re-estimate the baseline regression results. Dynamic treatment effect plots obtained from these four methods are presented in Fig. 4. The regression results from the four methods that correct for heterogeneous treatment effects indicate that there was no significant difference in carbon emission intensity between the treatment and control groups prior to the implementation of ETS. However, following the introduction of the ETS policy, the treatment group's carbon emission intensity began to decline significantly relative to the control group. This finding is consistent with the baseline regression results, confirming that the implementation of the carbon trading market has a progressively suppressive effect on carbon emission intensity in the treatment group over time.

The Bacon decomposition method proposed by Goodman-Bacon [69] decomposes the two-way fixed effect estimation results without adding control variables, and divides the total DID estimation effect into three comparison groups according to the group size and the variance of the treatment variable. (1) early adopters of the ETS versus late adopters, (2) late adopters versus early adopters, and (3) ETS adopters versus countries that never implemented the policy. The decomposition results are presented in Table 3. The total estimated coefficient from the Bacon decomposition without control variables is consistent with the baseline regression estimate without control variables (-0.068), suggesting that the coefficient of the core explanatory variable remains robust to treatment effect heterogeneity. Moreover, all three comparison groups yield negative estimated coefficients, indicating that potential heterogeneity does not materially affect the reliability of the overall estimate. These findings further confirm that the implementation of ETS policies is associated with a significant downward trend in carbon emission intensity in the adopting countries.

4.5. Robustness checks

4.5.1. Replacing the dependent variable

To further assess the robustness of the ETS policy's emission reduction effect on carbon intensity, we re-estimate the baseline model by substituting the dependent variable with carbon emissions per capita. Column (1) of Table 4 presents the results of this robustness check. The estimated coefficient of the ETS policy on carbon emissions per capita is -0.137 , which is statistically significant at the 1% level. This suggests

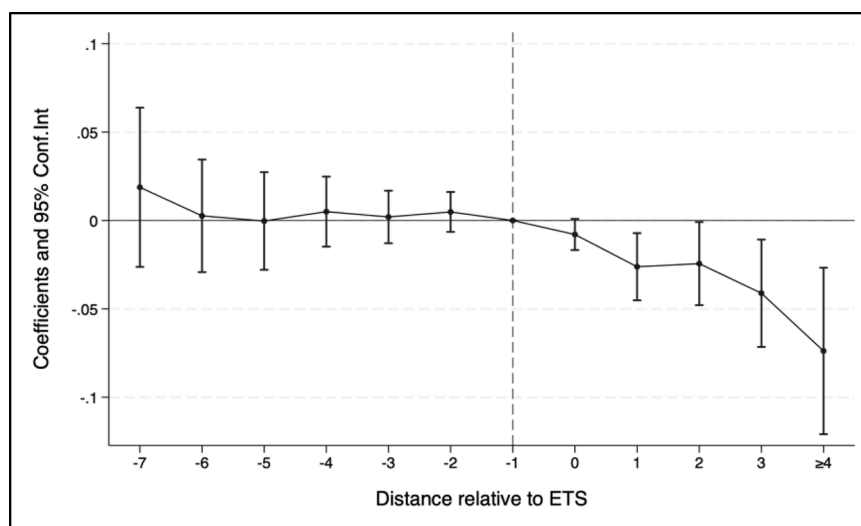


Fig. 2. Parallel Trend Test.

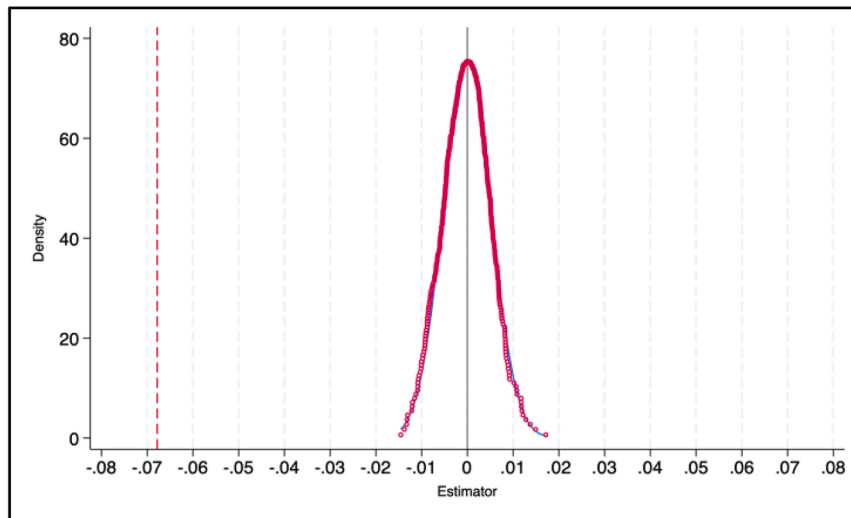


Fig. 3. Placebo Test.

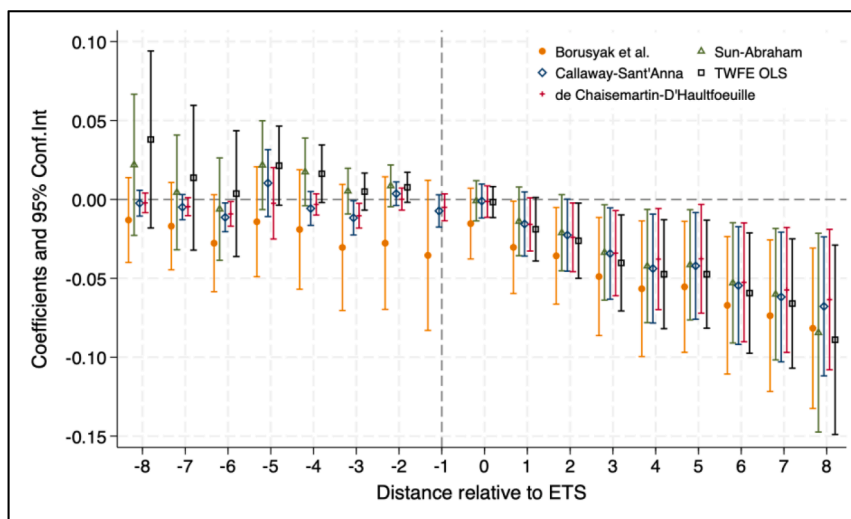


Fig. 4. Heterogeneous Treatment Effects.

Table 3
Results of Bacon Decomposition Test.

Category	Average DID Estimate	Weight
Early Treated vs. Late Treated	−0.048	0.069
Late Treated vs. Early Treated	−0.024	0.059
Treated vs. Never Treated	−0.075	0.872
DID Total Estimate	−0.068*** (−2.63)	

that the implementation of ETS policies significantly reduces carbon emissions per capita in participating countries, a finding that is consistent with the baseline regression results.

4.5.2. Controlling for joint fixed effects

To account for potential omitted variable bias arising from unobserved interactions between country-specific and time-specific factors, we follow Liu & Xu [70] and augment the baseline model with country-time interaction fixed effects. This approach allows us to capture the differentiated impact of common factors on different countries and across different time periods. As reported in Column (2) of Table 4, the coefficient on the ETS remains significantly negative after

introducing the interaction fixed effects, thereby reinforcing the robustness and validity of the main findings.

4.5.3. Entropy balancing test

In this study, countries implementing carbon emissions trading markets are designated as the treatment group, while those without such markets serve as the control group. Given the potential for differences in covariates between the treatment and control groups, and the possibility that these differences are directly related to carbon intensity, we employ Entropy Balancing—a method known for its accuracy in handling high-dimensional data—to enhance the credibility of our baseline regression. This approach simultaneously controls the first-order, second-order cross-moments, and third-order moments of the covariates for both the treatment and control groups, seeking to achieve accurate matching of the two groups' sample covariates while satisfying imposed constraints to the greatest extent possible. We then utilize Logistic regression, while controlling for balanced regional distribution, to estimate the Entropy Balancing weights. The regression results of the ETS policy on carbon intensity after Entropy Balancing are presented in Column (3) of Table 4. The results show that even after Entropy Balancing, the implementation of ETS policies continues to significantly (at the 1% significance level)

Table 4

Results of Robustness tests.

	Replacing the Dependent Variable (1)	Controlling for Joint Fixed Effects (2)	Entropy Balancing Test (3)	Random Forest (4)	Lasso Regression (5)	Anticipation effect (6)
ETS	CEA_{it} −0.137*** (−3.50)	CEI_{it} −0.054** (−2.33)	CEI_{it} −0.052*** (−3.17)	CEI_{it} −0.050*** (−4.48)	CEI_{it} −0.052*** (−4.73)	CEI_{it} −0.06,320*** (−4.34)
ETS_Pre1						−0.01,095 (−0.67)
ETS_Pre2						0.00,353 (0.30)
Control	YES	YES	YES	YES	YES	YES
Countryfixed	YES	YES	YES	YES	YES	YES
Timefixed	YES	YES	YES	YES	YES	YES
Time*Country fixed	NO	YES	NO	NO	NO	NO
N	2328	2328	2328	2328	2328	2328
R ²	0.960	0.859	0.928	—	—	0.859

suppress the increase in carbon emission intensity, remaining consistent with the baseline regression findings.

4.5.4. Random forest and lasso regression

To address the limitations of traditional DID model—such as the curse of dimensionality, the difficulty of exhaustively controlling for key covariates, and the inability of linear combinations of control variables to accurately fit the regression model—this study incorporates a double machine learning framework within the quasi-natural experiment to conduct causal inference on the impact of ETS implementation on carbon emission intensity. The sample is split in a 1:2 ratio, and robustness checks are conducted using both the Random Forest algorithm and the Lasso regression method. As reported in Columns 4 and 5 of Table 4, the estimated coefficients for ETS implementation remain significantly negative at the 1% level, consistent with the baseline regression results, thereby reinforcing the validity of the findings.

4.5.5. Anticipation effects

Considering that countries implementing ETS at an early stage may generate policy demonstration and expectation effects on countries that have not yet adopted ETS, governments may anticipate the introduction of ETS policies and adjust their emission reduction strategies in advance. Such anticipatory behaviors could potentially bias the estimated treatment effects and affect the validity of the main findings. This paper advances the timing of ETS implementation by one and two periods and constructs corresponding lead policy dummy variables to examine whether anticipation effects exist. The regression results are reported in the “Anticipation Effect” column of Table 4. The results show that the estimated coefficient of ETS variable remains significantly negative, consistent with the baseline regression and other robustness checks. Meanwhile, neither of the two lead policy variables designed to capture anticipation effects exhibits a significantly negative coefficient. These findings suggest that the implementation of ETS does not generate significant anticipation effects, and countries do not undertake substantial emission reduction actions merely based on expectations of future ETS adoption. Therefore, the estimated negative impact of ETS on carbon emission intensity reflects a genuine treatment effect after policy implementation rather than a spurious relationship driven by pre-policy behavioral adjustments in anticipation of ETS adoption.

4.6. Mechanism analysis

4.6.1. Mediation effects

Energy Consumption Structure. The results presented in column 1 of Table 5 indicate that the implementation of ETS significantly increases the share of renewable energy generation in the treated countries. The rising proportion of renewable energy sources—such as wind and photovoltaic power—substitutes for conventional fossil fuel-based

Table 5

Results of Mediation Effects.

	ECS (1)	GTI (2)
ETS	6.625*** (4.09)	4.007* (1.91)
Control	YES	YES
Country fixed	YES	YES
Time fixed	YES	YES
N	2138	1942
R ²	0.900	0.885

power generation (e.g., coal and oil), thereby promoting a transition in the energy structure from fossil dependence toward a more diversified and cleaner energy mix. This structural shift contributes to lowering carbon emission intensity, enhancing energy efficiency, and promoting sustainability. Consequently, the increased share of renewable energy generation plays a crucial role in reducing both total carbon emissions and emission intensity. These findings suggest that ETS helps mitigate carbon emission intensity by optimizing the energy consumption structure.

2) Green Technology Innovation. The results reported in column 2 of Table 5 demonstrate that the implementation of ETS significantly enhances the level of green technological innovation in the treated countries. On the one hand, ETS stimulates high-emission industries to increase investment in research and development (R&D) and to attract scientific and technical personnel, thereby strengthening their green innovation capacity. On the other hand, the penalty mechanism under the ETS compels countries with excessive carbon emissions to adopt greener production modes. This, in turn, facilitates increased public and private support for R&D and talent acquisition, fostering a positive feedback loop that promotes sustained green innovation and carbon mitigation. Accordingly, the implementation of ETS reduces carbon emission intensity by promoting green technological innovation. Hypothesis 1 is totally supported.

4.6.2. Moderation effects

Column (1) of Table 6 reports a significantly negative coefficient for the interaction term between ETS and FIN, suggesting that a higher level of financial development strengthens the emission-reduction effect of ETS. A more developed financial system can enhance the efficiency of carbon markets by improving capital allocation, facilitating investment in low-carbon technologies, and strengthening the transmission of carbon price signals. Moreover, financial development can increase market liquidity and improve carbon pricing mechanisms, thereby encouraging firms and other economic agents to undertake greater emission reduction efforts. These channels collectively amplify the effectiveness of ETS in reducing carbon emission intensity.

Similarly, Columns (2) and (3) show that the interaction terms

Table 6
Results of Moderation Effects.

	FIN	Carbon Market Maturity	
		C_PRICE	C_RATIO
	CEI_{it} (1)	CEI_{it} (2)	CEI_{it} (3)
ETS	0.041 (1.17)	−0.029 (−1.51)	−0.012 (−1.12)
FIN	0.004 (1.14)		
ETS*FIN	−0.011** (−2.38)		
C_PRICE		0.000 (0.00)	
ETS*C_PRICE		−0.001* (−1.71)	
C_RATIO			2.529** (2.57)
ETS*C_RATIO			−1.348*** (−2.70)
Control	YES	YES	YES
Country fixed	YES	YES	YES
Time fixed	YES	YES	YES
N	2035	2121	1993
R ²	0.870	0.846	0.846

between ETS and carbon price, as well as between ETS and emission coverage, are both significantly negative. This result indicates that ETS generates stronger carbon intensity reduction effects in economies with more mature carbon markets, characterized by higher carbon prices and broader emission coverage. The reason lies in that a mature carbon market is equipped with more refined market trading mechanisms and effective price signals. This compels firms to more actively seek pathways to reduce carbon emissions, consequently strengthening the inhibitory effect of ETS on carbon emission intensity in a country. Taken together, these findings provide strong support for Hypothesis 3.

4.7. Heterogeneity tests

4.7.1. Natural geographic factors

Regarding the cumulative factor, column 1 of Table 7 shows that the coefficient of ETS is significantly negative for countries with a higher proportion of low-altitude land areas (above the median value), indicating that the cumulative factor strengthens the emission reduction effect of ETS. Low-altitude areas possess significant carbon storage capacity, as vegetation coverage and sedimentation processes contribute to long-term carbon accumulation. Meanwhile, relatively weaker topographic barriers in these areas facilitate atmospheric circulation and pollutant dispersion, thereby enhancing the effectiveness of ETS in reducing carbon emission intensity.

Regarding the carbon sequestration factor, column 2 of Table 7 indicates that the emission reduction effect of ETS is stronger in countries with higher levels of natural resource dependence. These economies generally rely heavily on natural resources, particularly fossil fuels, whose extraction and utilization are associated with substantial carbon

emissions. Since ETS in most countries primarily regulate carbon-intensive industries by imposing emission caps and increasing the cost of excessive emissions, resource-dependent countries tend to experience stronger incentives for emission reductions after the implementation of ETS. Consequently, ETS generates a more pronounced effect on reducing carbon emission intensity in countries with greater dependence on natural resources.

Regarding the deposition factor, column 3 of Table 7 demonstrates that countries with annual precipitation depth above the median value exhibit stronger emission reduction effects of ETS. This result may be attributed to the fact that sufficient precipitation provides favorable water conditions for vegetation growth, enhances ecosystem vitality, and improves natural carbon sink capacity. By strengthening the absorption of anthropogenic carbon emissions through natural ecosystems, higher precipitation levels further reinforce the emission reduction effectiveness of ETS.

4.7.2. Economic development factors

Table 8 presents the heterogeneous effects of ETS on carbon emission intensity across economies with different levels of economic development. Following the World Bank's 2025 income classification based on per capita gross national income (GNI), economies are categorized into two groups: lower-middle-income economies (with per capita GNI ranging from 1136 to 4495 US dollars) and upper-middle-income economies (with per capita GNI exceeding 4496 US dollars). Accordingly, grouped regressions are estimated separately for the full sample and the subsample of developing economies. Columns (1) and (2) report the estimation results for the full sample. The coefficient of ETS is significantly negative for upper-middle-income economies, indicating that ETS implementation effectively reduces carbon emission intensity in these economies. However, the estimated coefficient for lower-middle-income economies is statistically insignificant. This finding suggests that the effectiveness of ETS may depend on the level of

Table 8
Heterogeneity Tests_Economic development Factors.

	Full Sample Economies		Developing Economies	
	Upper-middle-income Group	Lower-middle-income Group	Upper-middle-income Group	Lower-middle-income Group
ETS	−0.05,916*** (−2.79)	−0.09,458 (−1.66)	−0.07,993* (−1.95)	−0.09,458 (−1.66)
Chow test (P value)	4.85*** (0.000)		4.42*** (0.000)	
Control	YES	YES	YES	YES
Country fixed	YES	YES	YES	YES
Time fixed	YES	YES	YES	YES
N	1412	916	460	916
R ²	0.893	0.833	0.937	0.833

Table 7
Heterogeneity Tests_Natural Geographic Factors.

	HEI		RENT		APD	
	above-median group	below-median group	above-median group	below-median group	above-median group	below-median group
ETS	−0.070** (−2.86)	−0.027 (−0.64)	−0.117** (−2.61)	−0.043*** (−2.84)	−0.054** (−2.91)	−0.017 (−0.43)
Chow test	12.26*** (0.000)		7.18*** (0.000)		45.80*** (0.000)	
Fisher's Permutation (P value)	0.043** (0.014)		0.075*** (0.000)		0.038** (0.025)	
Control	YES	YES	YES	YES	YES	YES
Country fixed	YES	YES	YES	YES	YES	YES
Time fixed	YES	YES	YES	YES	YES	YES
N	951	985	1164	1164	1127	1132
R ²	0.813	0.856	0.859	0.902	0.877	0.871

economic development, as upper-middle-income economies generally possess more mature environmental regulatory systems, stronger institutional capacity, and more developed market mechanisms, which together enhance the effectiveness of market-based carbon regulation. Columns (3) and (4) illustrate the emission reduction effects of ETS among developing economies. The results show that the ETS coefficient is significantly negative for upper-middle-income developing economies, while it is insignificant for lower-middle-income developing economies. One possible explanation can be linked to the Environmental Kuznets Curve. As economies move toward higher income levels, they typically exhibit greater environmental awareness, stronger demand for environmental quality, more stringent environmental regulations, and enhanced capacity for low-carbon transformation. These factors provide a more favorable institutional and market environment for carbon markets to generate emission reduction effects. In contrast, although lower-middle-income developing economies have established carbon markets, their market mechanisms remain relatively underdeveloped. The operation of ETS relies heavily on administrative intervention rather than market-based adjustment [21]. Carbon allowances are mostly allocated free of charge alongside low carbon prices, which ultimately leads to insignificant emission reduction effects Table 9.

5. Extended analysis

The previous analysis based on the multi-period difference-in-differences (DID) model demonstrates that ETS implementation reduces carbon emission intensity in adopting countries by improving energy consumption structures and promoting green technological innovation. Moreover, the emission reduction effects are more pronounced in countries with more mature carbon markets. Meng et al. [54] find that narrowing the gap in emission intensities between developed and developing countries can mitigate carbon leakage associated with global trade. However, differences in the development stages of carbon markets across countries may lead to heterogeneous reductions in carbon emission intensity. Increasing disparities in sectoral carbon emission intensities among countries may further encourage the relocation of carbon-intensive production activities and exacerbate global carbon leakage, where emission reductions in one country are offset by emission increases elsewhere. Given this concern, the following analysis estimates global carbon flows and further examines whether ETS implementation affects consumption-based carbon emissions, consumption-based carbon emission intensity, and net carbon transfers, and then identifies the causal relationship between ETS adoption and global carbon transfers. This analysis provides important implications for accelerating the establishment of an integrated global carbon market and mitigating potential increases in cross-border carbon leakage.

Table 9
Average Carbon Emission Intensity and Its Trends by Country Group (2000–2020)³.

Category	Average Carbon intensity 2000 (Kt/\$M)	Average Carbon intensity 2020 (Kt/\$M)	Variation in Average Carbon Intensity between 2000 and 2020
Exceed target (3 Countries)	0.46	0.13	−71.9%
Meet target (7 Countries)	0.75	0.27	−63.5%
Below target (26 Countries)	1.90	0.49	−74.4%
No carbon market (32 Countries)	2.09	0.80	−61.5%

5.1. Describing statistics of global “carbon leakage”

The main method of carbon emission accounting in all countries is still based on the production-based approach. Enterprises in countries with strict environmental regulations tend to transfer the production and production processes of pollution-intensive products to those with lax environmental regulations. This practice reduces their own carbon emission responsibilities, thereby giving rise to the global issue of carbon leakage and hindering the achievement of global carbon reduction targets [71,72]. The World Bank's 2024 Carbon Pricing Report shows that only 6 countries worldwide have a carbon market price exceeding USD 100 per ton, with these primarily concentrated in Northern Europe. Most countries have a carbon price below USD 40 per ton, indicating inadequate efficiency of carbon markets. The significant carbon price difference reflects the variation in environmental costs across countries, which may further exacerbate the global problem of carbon leakage. In view of this, based on the industry-specific carbon emission data of various countries provided by the OECD Inter-Country Input-Output Tables (ICIO) and the Emissions Database for Global Atmospheric Research (version v8.0_FT_2022) ¹ this study matches industries² and then uses the MRIO model [73] to calculate the production-based carbon emissions, consumption-based carbon emissions, and net carbon transfer of 69 countries (Appendix 2 provides the calculation method). The results are presented in Fig. 5.

In Fig. 5, between 2000 and 2020, according to the World Bank's carbon market classification, countries including Switzerland, Sweden, and Finland (with a carbon price exceeding USD 100 per ton, categorized as “Exceeds Target” and grouped under the “EU_E” in the figure), as well as Canada, Germany, France, the United Kingdom, Austria, Ireland, and the Netherlands (with a carbon price exceeding USD 45 per ton, categorized as “Meets Target” and grouped under the “EU_M” in the figure), all exhibited net carbon emission outflows (represented by negative values in the bar chart). Among countries with a carbon price below USD 40 per ton (labelled with Below Target), OECD countries such as the United States, Japan, South Korea, and EU_B (low-carbon-price European countries, including Belgium, Bulgaria, the Czech Republic, Denmark, Spain, Estonia, Greece, Croatia, Hungary, Italy, Latvia, Lithuania, Poland, Portugal, Romania, Slovakia, and Slovenia) generally showed net carbon outflows—of which the United States accounted for 36% of the global total net carbon outflows in 2020. In contrast, developing economies like China and Kazakhstan exhibited net carbon inflows over the years, with China accounting for 45% of the global total net carbon inflows. In summary, economically developed countries with mature carbon markets have stricter domestic environmental regulations. Their enterprises tend to transfer polluting industries and products to other regions with lower environmental costs through international trade and investment. Such differentiation in the global carbon market may lead to carbon leakage.

Furthermore, based on the World Bank's classification of carbon markets, countries classified as “exceed target” exhibit the lowest average carbon emission intensity during 2000–2020, whereas countries without carbon markets show the highest average carbon emission intensity. The average carbon emission intensity of “meet target” and “below target” countries falls between these two groups. Compared with 2000, all four categories of countries experienced substantial declines in carbon emission intensity by 2020. Specifically, the average carbon emission intensity of exceed-target countries declined from approximately 0.46 Kt/\$M in 2000 to 0.13 Kt/\$M in 2020, representing a reduction of 71.9%. Meanwhile, countries without carbon markets

¹ Data download link: <https://data.jrc.ec.europa.eu/dataset/809d7b72-55ef-4e52-8bd4-7d33f2f9916b>.

² The 8 industries after matching are as follows: Agriculture, Energy Extraction, Industrial Combustion, Processing and Manufacturing, Energy Supply, Construction, Transportation, and Commercial and Public Services.

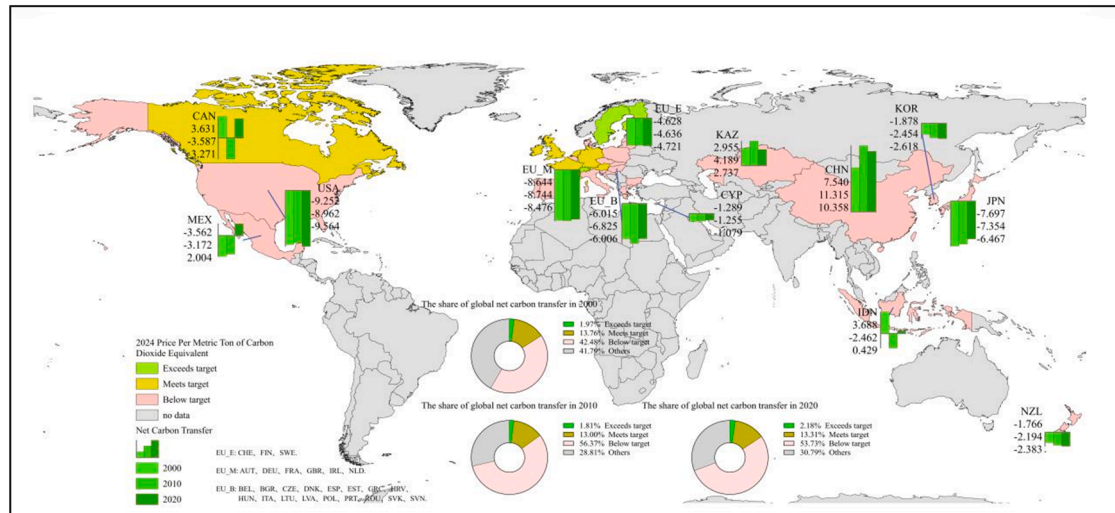


Fig. 5. Carbon price distribution and net carbon transfer in major global carbon markets (2000–2020). For visual clarity, all values were cube-rooted.

experienced a decline of 61.5% over the same period. Overall, countries implementing ETS achieved a larger reduction in average carbon emission intensity compared with countries without ETS. These descriptive statistics suggest that ETS effectively reduce carbon emission intensity in adopting countries, with a greater magnitude of decline relative to non-adopting countries. However, the differential improvement in carbon emission intensity associated with ETS implementation may widen the gap in carbon emission intensity across countries. Such increasing disparities may create incentives for the relocation of carbon-intensive production activities and potentially contribute to greater global carbon transfers. Therefore, the following analysis employs econometric approaches to further identify the causal relationship between ETS implementation and cross-border carbon transfers.

5.2. ETS and global “carbon leakage”

This study further calculates the ratio of the absolute value of net carbon transfers to production-based carbon emissions for each country. A higher value of this ratio represents a greater degree of carbon leakage in the corresponding country. In addition, since consumption-based carbon emissions reflect the carbon footprint embodied in a country’s final demand, countries with higher consumption-based emissions generally exhibit greater carbon outflows. Therefore, consumption-based carbon emissions (lnCBA) and consumption-based carbon emission intensity (lnCBAI) are also appropriate indicators for capturing cross-border carbon transfers. By re-estimating the multi-period DID model with ETS as the treatment variable and the three carbon leakage indicators as dependent variables, we find that ETS significantly increases the net carbon transfer ratio, consumption-based carbon emissions, and consumption-based carbon emission intensity in adopting countries (Table 10). The introduction of carbon trading mechanisms imposes stricter carbon constraints and increases emission reduction

costs for carbon-intensive firms within regulated jurisdictions. To avoid rising compliance costs associated with local environmental regulations, firms may relocate the production of emission-intensive goods and related production processes to regions with relatively lower carbon constraints. Although pollutant emissions within the local territory seem to decrease, this essentially creates an “emission reduction illusion”. Specifically, carbon emissions embodied in production activities do not disappear but are instead transferred to other regions through international trade and investment channels, thereby intensifying global carbon leakage. The trade-off effect of ETS has trapped global climate governance in a dilemma of regulatory arbitrage. Therefore, strengthening international carbon market linkages, promoting the development of an integrated global carbon market, and reducing cross-country disparities in carbon prices are effective approaches to reducing carbon emission intensity and preventing carbon leakage.

6. Discussion

Although many countries have made efforts to combat global warming, however the overall temperatures are still increasing worldwide and realize the climate objectives of the Paris Agreement are still facing huge challenge. We document the important role of the ETS in reducing carbon emissions, and the underlying reduction mechanisms (i. e., optimizing the energy consumption structure and incentivizing green technological innovation). The results also indicates that financial development and carbon market maturity will significantly amplify the emission-reduction impact of ETS. The trade-off effect induced by ETS will undermine global mitigation efforts and the achievement of global mitigation objectives. Therefore it is important to strengthen global carbon market links and control the carbon leakage of ETS. First, to strengthen the links between carbon markets in various countries and reduce carbon price differences, it is necessary to formulate unified cross-border carbon emission monitoring standards. New digital technologies should be employed to establish unified international MRV standards for carbon emissions in order to transform carbon allowances and voluntary emission reductions—whether as Real World Assets (RWA) or Real Data Assets (RDA)—into divisible and tradable on-chain tokens. This would facilitate the development of a decentralized cross-border carbon trading mechanism. Second, a real-time global carbon flow tracking platform should be established to facilitate data sharing, enabling the timely identification and early warning of carbon-intensive industries and trade flows at high risk of carbon leakage. Third, under the framework of the principle of “common but differentiated responsibilities,” a Global Cross-Border Carbon Leakage Compensation

Table 10
Regression results of the effect of ETS on carbon leakage.

	NCT_Ratio (1)	lnCBAI (2)	lnCBA (3)
ETS	5.499* (1.82)	0.359** (2.22)	0.400** (2.24)
Control	YES	YES	YES
Country fixed	YES	YES	YES
Time fixed	YES	YES	YES
N	1176	1176	1176
R ²	0.868	0.173	0.173

Fund should be established by UNFCCC. This fund would channel low-carbon technology subsidies from high carbon-price jurisdictions to regions adversely affected by industrial relocation driven by carbon price differentials. Such a mechanism would help prevent the unilateral transfer of mitigation costs to developing countries and support the equitable and coordinated development of the global carbon market. All in all, this paper has broad policy implications in all countries worldwide and prompt to set up a unified global carbon market.

This study has several limitations that warrant further investigation. First, there are approximately 130 globally traceable climate governance policies (CAPMF), yet this analysis does not comprehensively consider their interactive effects. Our research primarily focuses on identifying the emission reduction effects and mechanisms of market-based ETS, without fully accounting for the combined impact of other major climate governance policies and their interaction with ETS policies. In practice, achieving global climate governance objectives cannot rely on a single policy instrument alone; rather, it requires the development of effective and coordinated policy portfolios. Future research should therefore employ more comprehensive and systematic modeling approaches to identify optimal policy mixes and evaluate their joint effectiveness. Furthermore, while ETS has proven highly effective in reducing domestic carbon emission intensity, carbon leakage—driven by the disparities across national carbon markets—persists as a long-standing issue in the absence of a globally unified carbon market. Although the European Union has proposed the CBAM to mitigate the carbon leakage, it has encountered considerable opposition from many developing countries, which have raised concerns regarding its fairness and potential implications for international trade. Therefore, achieving a balance between the emission reduction benefits of ETS and their potential trade-related spillover effects requires more comprehensive institutional arrangements and coordinated governance frameworks from multiple dimensions.

CRedit authorship contribution statement

Wenzhi Wang: Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Sijia Fan:** Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Data curation. **Yong Hu:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work is supported by the the Humanities and Social Sciences Planning Fund of the Ministry of Education of China (Grant No 25YJA790068), National Social Science Fund of China (Grant No 25BGJ106), and Tianjin Normal University Postgraduate Research Innovation Project (Grant No 2026KYCX014Z)

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.sfr.2026.102156](https://doi.org/10.1016/j.sfr.2026.102156).

Data availability

Data will be made available on request.

References

- [1] A. Stechemesser, N. Koch, E. Mark, E. Dilger, P. Klösel, L. Menicacci, D. Nachtigall, F. Pretis, N. Ritter, M. Schwarz, Climate policies that achieved major emission reductions: global evidence from two decades, *Sci.* (1979) 385 (2024) 884–892.
- [2] R. Schmalensee, R.N. Stavins, Lessons learned from three decades of experience with cap and trade, *Rev. Environ. Econ. Policy.* 11 (1) (2017) 59–79.
- [3] Y. Chen, J. Miao, Z. Zhu, Measuring green total factor productivity of China's agricultural sector: a three-stage SBM-DEA model with non-point source pollution and CO₂ emissions, *J. Clean. Prod.* 318 (2021) 128543.
- [4] M. Lu, X.N. Fang, Game analysis on carbon market from allocation of allowance, *Chin. J. Manag. Sci.* 23 (S1) (2015) 807–811.
- [5] R. Betz, M. Sato, Emissions trading: lessons learnt from the 1st phase of the EU ETS and prospects for the 2nd phase, *Clim. Policy.* 6 (4) (2006) 351–359.
- [6] S.M. Bartram, K. Hou, S. Kim, Real effects of climate policy: financial constraints and spillovers, *J. financ. econ.* 143 (2) (2022) 668–696.
- [7] S. Li, J. Liu, D. Shi, The impact of emissions trading system on corporate energy efficiency: evidence from a quasi-natural experiment in China, *Energy* 233 (2021) 121129.
- [8] M. Song, H. Zheng, Z. Shen, Whether the carbon emissions trading system improves energy efficiency – Empirical testing based on China's provincial panel data, *Energy* 275 (2023) 127465.
- [9] R. Wu, Z. Tan, B. Lin, Does carbon emission trading scheme really improve the CO₂ emission efficiency? Evidence from China's iron and steel industry, *Energy* 277 (2023) 127743.
- [10] Y. Zhao, Y. Xu, Carbon emission trading policy, information disclosure, and effects of pollution reduction and carbon mitigation: empirical evidence from China, *Sustain. Futures* 11 (2026) 101637.
- [11] J.S. Wang, M.L. Gong, Z.S. Wang, Z.Y. Wei, Y. Liu, The study of enterprise response to carbon trading market based on agent-based modeling, *Chin. J. Environ. Manage.* 16 (5) (2024) 138–146.
- [12] B. Li, X.Q. Xia, Q.Y. Li, Research on the influence of carbon trading on carbon emission reduction effect and coordination mechanisms under carbon quotas, *Chin. J. Manag. Sci.* 32 (8) (2024) 250–261.
- [13] Y. Wu, X.X. Wang, Improvement of China's carbon Quota Allocation Mechanism Under the Perspective of Equity and Efficiency, 52, *Environmental Protection*, 2024, pp. 24–27.
- [14] X.X. Zhang, S.J. Sun, Can carbon emission trading policy promote carbon emission reduction: a study based on prefecture-level city panel data, *Nankai Economic Stud.* (2) (2024) 160–178.
- [15] X.W. Xuan, H. Zhang, International experiences and lessons of carbon emission permit allocation method, *Chin. J. Popul. Resour. Environ.* 23 (12) (2013) 10–15.
- [16] J.R. Zhang, Z.D. Wang, L. Tang, L. Yu, The simulation of carbon emission trading system in Beijing-Tianjin-Hebei region: an analysis based on system dynamics, *Chin. J. Manag. Sci.* 24 (3) (2016) 1–8.
- [17] D.D. Wang, B. Yang, Driving mechanism of carbon emissions trading system on green technology innovation of emission-controlled enterprises: from the perspective of market logic, *Soft Sci.* 38 (12) (2024) 71–78.
- [18] Z. Chen, X. Zhang, F. Chen, Do carbon emission trading schemes stimulate green innovation in enterprises? Evidence from China, *Technol. Forecast. Soc. Change* 168 (2021) 120744.
- [19] Z.Q. Dong, H. Wang, Validation of market-based environmental policies: empirical evidence from the perspective of carbon emission trading policies, *Stat. Res.* 38 (10) (2021) 48–61.
- [20] D. Fan, J.W. Fu, W.G. Wang, How does carbon emission trading influence firm's total factor productivity, *Syst. Eng. Theory Pract.* 42 (3) (2022) 591–603.
- [21] Y.Y. Wu, J. Qi, Q. Xian, J.D. Chen, The carbon emission reduction effects of China's carbon market—From the perspective of the coordination between market mechanism and administrative intervention, *China Ind. Econ.* (8) (2021) 114–132.
- [22] S.Y. Chen, Low-carbon economy, *J. Econ. Res.* 57 (6) (2022) 12–18.
- [23] C. Jeffrey, J.D. Perkins, The association between energy taxation, participation in an emissions trading system, and the intensity of carbon dioxide emissions in the European Union, *Int. J. Account.* 50 (4) (2015) 397–417.
- [24] Q. Ma, G. Yan, X. Ren, X. Ren, Can China's carbon emissions trading scheme achieve a double dividend? *Environ. Sci. Pollut. Res.* 29 (2022) 50238–50255.
- [25] Y.F. Hu, Y.Q. Ding, Can carbon emission permit trade mechanism bring both business benefits and green efficiency? *China Popul. Resour. Environ.* 30 (3) (2020) 56–64.
- [26] Z.G. Tu, R.J. Shen, Can emissions trading schemes achieve the Porter hypothesis in China? *J. Econ. Res.* 50 (7) (2015) 160–173.
- [27] H.T. Shen, N. Huang, L. Liu, A study of the micro-effect and mechanism of the carbon emission trading scheme, *J. Xiamen Univ. (Arts Soc. Sci.)* (1) (2017) 13–22.
- [28] X. Shi, C. Sun, A preliminary study on the legal system construction of "One Belt and One Road" carbon market, *Theor. Mod.* 2 (2020) 79–90.
- [29] H.T. Shen, N. Huang, Will the carbon emission trading scheme improve firm value? *Financ. Trade Econ.* 40 (1) (2019) 144–161.
- [30] M.S. Duan, Coordinations between China's emissions trading system and policies of energy efficiency and renewable energy, *J. Environ. Econ.* 3 (2) (2018) 1–10.
- [31] H.D. Jiang, L.J. Liu, K.Y. Dong, Y.W. Fu, How will sectoral coverage in the carbon trading system affect the total oil consumption in China? A CGE-based analysis, *Energy Econ.* 110 (2022) 105996.
- [32] M.M. Zhang, Y.Q. Ge, L.Y. Liu, D.Q. Zhou, Impacts of carbon emission trading schemes on the development of renewable energy in China: spatial spillover and mediation paths, *Sustain. Prod. Consum.* 32 (2022) 306–317.
- [33] X. Ren, Y. Xiao, K. Duan, Spillover effects between fossil energy and green market ETS: evidence from informational inefficiency, *Energy Econ.* 131 (2024) 107317.

- [34] J. Bai, H. Ru, Carbon emissions trading and environmental protection: international evidence, *Manage Sci.* 70 (7) (2024) 4593–4603.
- [35] J. Lv, Y.Q. Lv, P. Yang, J. Chen, Whether carbon emissions trading policy promotes "quantity and quality upgrading" of green technological innovations by firms? *Ecol. Econ.* 41 (3) (2025) 75–85.
- [36] P. Kuang, R.R. Xie, S.G. Liang, Pollution abatement and carbon reduction effects of carbon emission trading pilot policies—Empirical evidence based on multi-period double-difference modeling, *Rev. Econ. Manag.* 41 (6) (2025) 148–160.
- [37] E. Uyarra, P. Shapira, A. Harding, Low carbon innovation and enterprise growth in the UK: challenges of a place-blind policy mix, *Technol. Forecast. Soc. Change* 103 (2016) 264–272.
- [38] X.X. Yang, Y.W. Zhang, Carbon emission reduction effect and mechanism test of carbon emission trading pilot, *Ecol. Econ.* 19 (4) (2023) 302–316.
- [39] F. Li, Z.X. Weng, Spatial spillover effects of a carbon trading policy on green technology innovation in China, *Environ. Sci. Pollut. Res.* 30 (50) (2023) 109500–109524.
- [40] Z. Liu, H. Sun, Assessing the impact of emissions trading scheme on low-carbon technological innovation: evidence from China, *Environ. Impact. Assess. Rev.* 89 (2021) 106589.
- [41] R. Martin, M. Muûls, U. Wagner, The impact of the European Union emissions trading scheme on regulated firms: what is the evidence after ten years? *Rev. Environ. Econ. Policy.* 10 (2016) 129–148.
- [42] Y. Wei, J. Zhang, Y. Li, M. Wojewodzki, Z. Mou, M. Guo, Impact of emissions trading schemes on enterprise technological innovation: a scientometric review and future directions, *Sustain. Futures* 10 (2025) 101546.
- [43] M.E. Porter, C. van der Linde, Toward a new conception of the environment-competitiveness relationship, *J. Econ. Perspect.* 9 (4) (1995) 97–118.
- [44] S.Y. Jiao, Y. Yu, Carbon emission trading, enterprise green innovation and emission reduction incentives—From the perspective of corporate carbon emission reduction incentives, *Sci. Technol. Econ.* 6 (3) (2026) 75–80.
- [45] Q.Q. Fan, W. Guo, Impact of carbon trading market on firm innovation, *Res. Econ. Manag.* 46 (2) (2025) 43–57.
- [46] H.L. Jiang, W.D. Wang, L. Wang, J.H. Wu, The effects of the carbon emission reduction of China's green finance—An analysis based on green credit and green venture investment, *J. Mod. Finan.* 11 (2020), 39–48+80.
- [47] H.W. Niu, L.L. Cao, How does carbon price affect corporate green investment and credit risk: a trade-off analysis based on expected returns and default costs, *China Ind. Econ.* (8) (2024) 118–136.
- [48] S. Qi, L. Pang, T. Qi, The correlation between the green bond market and carbon trading market ETS under climate change: evidence from China, *Technol. Forecast. Soc. Change* 203 (2024) 123367.
- [49] Y.L. Wang, Y.H. Zhou, Green finance development and enterprise innovation, *J. Financ. Econ.* 49 (1) (2023) 49–62.
- [50] W. Dou, T.T. Li, W.Q. Zhang, Financial development and carbon emission intensity: a test based on the perspective of moderate level, *Environ. Econ. Res.* 8 (3) (2023) 132–155.
- [51] P.S. Lin, S. Li, Mandate-based and market-based policies: comparative analyses on carbon mitigation effectiveness of China's climate policies, *J. World Econ.* (6) (2024) 149–175.
- [52] D.B. Fang, Q.J. Xie, How carbon market impacts emission reduction in thermal power: from the perspective of carbon price, *Syst. Eng. Theory Pract.* (3) (2024) 1003–1017.
- [53] N. Döbbeling-Hildebrandt, K. Miersch, T.M. Khanna, et al., Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing, *Nat. Commun.* 15 (1) (2024) 41–47.
- [54] J. Meng, J. Huo, Z.W. Zhang, Z.F. Liu, Z.F. Mi, D.B. Guan, K.S. Feng, The narrowing gap in developed and developing country emission intensities reduces global trade's carbon leakage, 14 (2023) 37–75.
- [55] S. Shao, L.M. Ge, J.L. Zhu, How to achieve the harmony between humanity and nature: environmental regulation and environmental welfare performance from the perspective of geographical factors, *J. Manag. World* 40 (8) (2024) 119–146.
- [56] X.K. Xu, S.J. Zhang, Carbon emission reduction effects of global carbon markets: market mechanisms and government intervention, *Stud. Int. Financ.* (5) (2025) 29–39.
- [57] S. Kumar, H. Dhar, V.V. Nair, Environmental quality monitoring and impact assessment of solid waste dumpsites in high altitude sub-tropical regions, *J. Environ. Manage* 252 (2019) 109681.
- [58] N.H. Liu, Y.F. Wang, Y.J. Ma, Y. Hong, Simulation on influence of complex terrain on urban air pollution, *Geogr. Sci.* (3) (2008) 396–401.
- [59] K.Q. Li, Y. Li, M.H. Ou, How can land use enter the low-carbon era, *China Land*, (4) (2015) 10–12.
- [60] L. Qian, M.C. Liu, Y.L. Yang, G.H. Yan, X.B. Lan, Characteristics of change in solar radiation and solar energy resources use over the eastern Hexi Corridor, *Resour. Sci.* (5) (2011) 823–828.
- [61] T. Beck, R. Levine, A. Levkov, Big bad banks? The winners and losers from bank deregulation in the United States, *J. Financ.* 65 (5) (2010) 1637–1667.
- [62] B.L. Gong, Q.Z. Zhang, L.G. Yuan, X.G. Liu, The policy innovation and effect evaluation of the revitalization and development in old revolutionary base areas, *J. Manag. World* 38 (8) (2022) 26–43.
- [63] A.C. Baker, D.F. Larcker, C.C.Y. Wang, How much should we trust staggered difference-in-differences estimates? *J. financ. econ.* 144 (2) (2022) 370–395.
- [64] S. Athey, G.W. Imbens, Design-based analysis in difference-in-differences settings with staggered adoption, *J. Econom.* 226 (1) (2022) 62–79.
- [65] B. Callaway, P.H.C. Sant'Anna, Difference-in-differences with multiple time periods, *J. Econom.* 225 (2) (2021) 200–230.
- [66] C. De Chaisemartin, X. D'Haultfoeulle, Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: a survey, *Econom. J.* 26 (3) (2023) C1–C30.
- [67] K. Borusyak, X. Jaravel, J. Spiess, Revisiting event study designs: robust and efficient estimation, *Rev. Econ. Stud.* 91 (6) (2024) 3253–3285.
- [68] L. Sun, S. Abraham, Estimating dynamic treatment effects in event studies with heterogeneous treatment effects, *J. Econom.* 225 (2) (2021) 175–199.
- [69] A. Goodman-Bacon, Difference-in-differences with variation in treatment timing, *J. Econom.* 225 (2) (2021) 254–277.
- [70] Z.H. Liu, J.W. Xu, The influence and mechanism of carbon trading pilot on provincial carbon emissions equity: empirical analysis based on multi-period DID, spatial DID and intermediary effect, *J. Nat. Resour.* 39 (3) (2024) 697.
- [71] W.Z. Wang, J.W. Yao, The impact assessment of the EU carbon border adjustment mechanism on the export costs of carbon-intensive products in China: from the perspective of shared carbon emission responsibility, *Yuejiang Acad. J.* 16 (6) (2024), 66–77+179.
- [72] B. Zhou, C. Zhang, Q. Wang, D. Zhou, Does emission trading lead to carbon leakage in China? Direction and channel identifications, *Renew. Sustain. Energy Rev.* 132 (2020) 110090.
- [73] W.Z. Wang, Re-estimation of provincial consumption-side carbon emission responsibility allocation in China: from the perspective of shared responsibility and technological compensation, *Statistical Research* 39 (6) (2022) 3–16.